Ilya Venediktov

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EDUCATION

Ph.D. in Physics
Ohio University, Athens, OH, USA
Expected: May 2026
Dissertation: "Weak Gravitational Lensing and the LoVoCCS Survey: Empirical Bias Calibration with HST"
Advisor: Dr. Douglas Clowe
Focus: Weak Gravitational Lensing
GPA: 3.776/4

M.S. in Physics
Ohio University, Athens, OH, USA
Graduation: August 2022
GPA:3.735/4

B.S. in Astrophysics
Rutgers University, New Brunswick, NJ, USA
Dean's List (4 semesters)
Graduation: May
GPA: 3.333/4

A.S. in Physics
Brookdale Community College, Lincroft, NJ, USA
Dean's list (3 semesters), Phi Theta Kappa Honor Society, Distinguished Scholar Award
Graduation: May 2015
GPA: 3.81/4

RESEARCH EXPERIENCE

Research Assistant | *Department of Physics and Astronomy, Ohio University*
August 2022 – August 2025

- Prepared a dissertation prospectus focused on weak gravitational lensing shear bias calibration using HST and DECam data from the LoVoCCS survey
- Reviewed literature on shear systematics, including multiplicative bias, blending effects, and PSF-related distortions
- Practiced using analysis tools such as SExtractor, PSFEx, GalSim, and Astropy for shape measurement and catalog handling
- Began working with LSST Science Pipelines, including the customized run_steps pipeline developed for the LoVoCCS survey and related data processing workflows

Graduate Research Preparation | *Department of Physics & Astronomy, Ohio University*
May 2024 – April 2025

- Prepared a dissertation prospectus titled Weak Gravitational Lensing and the LoVoCCS Survey: Empirical Bias Calibration with HST
- Reviewed literature and outlined methodology for calibrating multiplicative shear bias using matched HST and DECam data
- Explored use of SExtractor, PSFEx, GalSim, and LSST Science Pipelines for weak lensing shape
- measurement and simulation
- Developed preliminary workflow for catalog cross-matching and shear comparison
- Gained experience in proposal writing, literature synthesis, and planning of observational cosmology research

TEACHING EXPERIENCE

Teaching Assistant | *Department of Physics and Astronomy, Ohio University*
August 2020– August 2022

- Instructed two lab sections of Introduction to Physics (PHYS 2001) each semester, engaging and supporting approximately 15 students per section through hands-on experiments and applied learning
- Held weekly office hours to provide one-on-one academic support, helping students strengthen their understanding of core physics concepts
- Evaluated lab reports and exams, delivering timely and constructive feedback to promote student progress and mastery of course objectives

Grader | *Department of Physics and Astronomy, Ohio University*
August 2021– December 2021

- Assessed and provided detailed feedback on weekly homework assignments for PHYS 4021: Introduction to Quantum Mechanics
- Reviewed problem sets involving wavefunctions, operators, eigenvalue problems, and the Schrödinger equation
- Ensured consistency in grading and supported instructors in evaluating student understanding of foundational quantum theory concepts

WORK EXPERIENCE

Assistant | *Imperium Construction Inc, Brooklyn, NY, USA*
May 2018 – August 2018

- Contributed to the preparation of detailed cost estimation sheets for public school construction contracts, focusing on accurate material and labor cost breakdowns.
- Conducted market research by contacting suppliers and manufacturers to obtain current pricing data for construction materials.
- Entered and organized cost data using spreadsheet software to support the estimation process and compiling bid proposals.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, Astropy, Matplotlib), C/C++, Bash, Perl, IDL, Jupyter Notebooks, Git
Data Analysis & Visualization: SExtractor, PSFEx, FITS file handling (Astropy, DS9), LaTeX
Astronomical Tools: Catalog cross-matching, GalSim (basic), LSST Science Pipelines (introductory), run_steps pipeline for LoVoCCS (introductory)

AWARDS & HONORS

Phi Theta Kappa Honor Society, Brookdale Community College
Distinguished Scholar Award, Brookdale Community College
Dean's List, Brookdale Community College (Spring 2014, Fall 2014, Spring 2015)
Dean's List, Rutgers University (Fall 2017, Spring 2018, Spring 2019, Spring 2020)

RELEVANT GRADUATE COURSEWORK

PHYS 5041 – Mathematical Methods in Physics I

- Covers complex variables, vector calculus, linear algebra, Fourier analysis, and differential equations.
- ***Learning outcomes:*** Solve physics problems using advanced mathematical techniques across classical and quantum contexts.

PHYS 6001 – Classical Mechanics

- Introduces Newtonian mechanics, Lagrangian and Hamiltonian formalisms, central force motion, and rigid body dynamics.
- ***Learning outcomes:*** Develop problem-solving skills in classical systems using variational principles and conservation laws.

PHYS 6020 – Quantum Mechanics I

- Focuses on wave mechanics, Schrödinger equation, operators, eigenvalue problems, and potential wells.
- ***Learning outcomes:*** Understand and solve basic quantum systems; interpret quantum postulates.

PHYS 6021 – Quantum Mechanics II
- Continues from Quantum I includes angular momentum, spin, perturbation theory, and identical particles.
- ***Learning outcomes:*** Apply approximation techniques to multi-dimensional and realistic quantum systems.

PHYS 6011 – Statistical Mechanics I
- Introduces microcanonical, canonical, and grand canonical ensembles, partition functions and thermodynamic quantities.
- ***Learning outcomes:*** Derive macroscopic properties from microscopic models using statistical methods.

PHYS 6031 – Electrodynamics I
- Covers electrostatics, magnetostatics, Maxwell's equations, boundary-value problems, and wave propagation.
- ***Learning outcomes:*** Solve classical electromagnetic problems using vector calculus and boundary conditions.

PHYS 5071 – Computer Simulation Methods in Physics
- Introduces numerical methods including Monte Carlo techniques and differential equation solvers.
- ***Learning outcomes:*** Develop and apply simulations to model physical systems computationally.

PHYS 6940 – Special Study
- Individual research projects in physics, conducted under faculty supervision.
- ***Learning outcomes:*** Gain experience in problem formulation, literature review, and data/code-driven research.

PHYS 8201 – Research Seminar: Astrophysics
- Graduate-level seminar presenting current research topics in astrophysics.
- ***Learning outcomes:*** Practice scientific presentation and critical reading of research literature.

ASTR 5271 – Observational Astrophysics
- Covers astronomical instrumentation, CCD imaging, photometry, and data reduction techniques.
- ***Learning outcomes:*** Conduct and analyze observations with research-grade instruments.

ASTR 5201 – Stellar Astrophysics and Radiation
- Covers stellar interiors, nuclear processes, radiative transfer, and spectral analysis.
- ***Learning outcomes:*** Understand physical principles governing stellar structure and evolution.

ASTR 5202 – Interstellar Medium and Galaxies
- Explores phases of the interstellar medium, star formation, galactic structure, and dynamics.
- ***Learning outcomes:*** Analyze how gas dynamics and stellar processes shape galaxies.

PHYS 7021 – Quantum Field Theory I (In Progress)
- Introduces quantum field theory, covering canonical quantization, scalar and spinor fields, and Feynman diagrams.
- ***Learning outcomes:*** Understand the quantization of fields and apply perturbative techniques to quantum interaction

RELEVANT UNDERGRADUATE COURSEWORK

Thermal Physics (01:750:351)
- Covers energy, entropy, the laws of thermodynamics, phase transitions, chemical equilibrium, and introduces statistical mechanics.
- ***Learning outcomes:*** Apply thermodynamic and statistical principles to physical and chemical systems.

Quantum and Atomic Physics (01:750:361)
- Introduces wave mechanics, the Schrödinger equation, hydrogen atom, angular momentum, and spin.
- ***Learning outcomes:*** Solve foundational quantum problems and describe atomic structure.

Mechanics II (01:750:382)
- Advanced study of Lagrangian and Hamiltonian mechanics, oscillations, and central force problems.
- ***Learning outcomes:*** Analyze motion in non-inertial frames and develop fluency with variational principles.

Electromagnetism I (01:750:385)
- Intermediate-level course covering electrostatics, electric potential, Laplace's equation, method of images, multipole expansion, electric fields in matter, polarization, dielectrics, magnetostatics, Biot– Savart law, Ampère's law, magnetic vector potential, and magnetized materials.
- ***Learning outcomes:*** Develop proficiency in solving boundary-value problems in electrostatics and magnetostatics; understand the behavior of electric and magnetic fields in various media.

Electromagnetism II (01:750:386)
- Continuation of Electromagnetism I, focusing on electrodynamics, including electromagnetic induction, Maxwell's equations, electromagnetic waves, energy and momentum conservation, scalar and vector potentials, radiation, and the relationship between electrodynamics and special relativity.
- ***Learning outcomes:*** Apply Maxwell's equations to dynamic systems; analyze electromagnetic wave propagation; understand the relativistic formulation of electromagnetism.

Principles of Astrophysics (01:750:341)
- Covers gravitational dynamics, binary systems, galaxy structure, gravitational lensing, relativity, and cosmology.
- ***Learning outcomes:*** Apply Newtonian gravity and relativity to astrophysical systems; model motion in one-, two-and N-body problems; understand black holes and cosmological principles

Principles of Astrophysics (01:750:342)
- Explores stellar structure, stellar evolution, nucleosynthesis, and the interstellar medium.
- ***Learning outcomes*:** Apply thermodynamics, nuclear physics, and radiative processes to model stars, interpret stellar lifecycles, and analyze physical conditions in galactic environments.

Observational Radio Astronomy (01:750:343)
- Introduces techniques for acquiring and analyzing radio frequency astronomical data.
- ***Learning outcomes:*** Operate radio telescopes and reduce observational datasets.

Observational Optical Astronomy (01:750:344)
- Covers photometry, CCD imaging, and optical data analysis.
- ***Learning outcomes:*** Conduct optical observations and extract scientific measurements.

Stars and Star Formation (01:750:441)
- Covers stellar physics: nuclear reactions, structure, evolution, and star-forming regions.
- ***Learning outcomes:*** Explain stellar lifecycle processes and compare theory to observations.

Galaxies and the Milky Way (01:750:443)
- Studies the structure, dynamics, and formation of galaxies.
- ***Learning outcomes:*** Interpret galactic observational data and understand large-scale evolution.

Introduction to Cosmology (01:750:444)
- Covers expanding universe, cosmic microwave background, and large-scale structure.
- ***Learning outcomes:*** Describe the standard cosmological model and interpret cosmological observations.

Computational Astrophysics (01:750:488 – Special Topics in Physics)
- Project-based course focused on computational techniques in astrophysics, including simulations and data analysis.
- ***Learning outcomes:*** Develop, implement, and apply numerical methods to analyze astrophysical phenomena and interpret results using real and simulated dataset.